Finite-sample edge-order inference and certified degree comparisons in correlation spanning trees

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Abstract

A correlation-tree hub can be identifiable before the entire tree is recovered, but projecting separate edge intervals can obscure this distinction. We construct simultaneous hub inference from direct edge-order tests. For independent observations, predetermined disjoint pairs reduce Kendall correlation comparisons to exact conditional McNemar tests; Holm adjustment gives finite-sample coverage, including discrete margins after independent tie randomization. Certified orders yield degree-difference bounds through graphic-matroid ranks. Cycle-path constraints strengthen a linear-programming relaxation with rationally certified integer endpoints, while the tree degree sum improves rank projection. We credit the classical components and distinguish projection validity from statistical calibration. Monte Carlo experiments demonstrate hub certification in separated models and expose substantial Pearson-bootstrap undercoverage. The independent-observation guarantee is therefore not extended to dependent returns. Reanalysis of 231 Polish equity windows is supplemented by 466 analyses varying lookback, liquid-stock universe and resampling blocks. Matched-universe degree widths improve by about 10–14% with longer samples, while all rank ranges remain full. Forward diagnostics distinguish persistent estimated edge orders from less persistent estimated hub identities without treating temporal agreement as population coverage.

Keywords: Kendall correlation, simultaneous inference, spanning tree, McNemar test, degree rank, partial order

1 Introduction

Which stock is a population hub of a correlation spanning tree? Answering this question requires more than repeatedly selecting the same vertex in bootstrap samples. Tree edges depend on comparisons between correlations, and degrees depend on several edges jointly. Separate uncertainty intervals for every edge can lose useful dependence information; separate degree intervals can lose the restrictions imposed by a common tree. This paper develops inference that retains comparisons at both stages.

Our statistical starting point is a direct, finite-sample comparison of two Kendall correlations. Predetermined disjoint observation pairs produce independent binary edge kernels. Their discordances lead to the classical exact McNemar test. Adjusting a fixed family of local comparisons by Holm's procedure yields simultaneous strict edge-order statements. Unlike a generic uniform deviation bound, this test conditions on the observed number of discordances. It also admits discrete margins through independent random tie breakers without changing the population Kendall tau-a target.

The second stage translates these statements into degree and rank bounds. We give explicit pairwise degree extrema on the graphic minor obtained from cut and cycle screening, add joint path exclusions to a linear-programming relaxation, and verify improved integer endpoints using rational dual certificates. Rank projection additionally enforces the tree identity $\sum_i d_i = 2p - 2$. These steps address different sources of conservatism: statistical calibration, loss of edge dependence, and loss of feasible degree configurations.

The ingredients have substantial precedents. Robust spanning trees and possible or necessary optimality under interval uncertainty are studied by Yaman et al. (2001); Kasperski and Zieliński (2007); Fortin et al. (2009). Zaffalon and Hutter (2005) already use dependence comparisons and partial orders to infer common tree structure. We do not claim the introduction of partial-order tree inference, the cut and cycle criteria, or the matroid greedy argument. Sensitivity analysis of minimum spanning trees is a further related algorithmic literature (Pettie 2014). Our contribution is the explicit Kendall-to-degree inference construction, its finite-sample guarantee under ties, the comparison and path-based projections with verifiable endpoints, and an evaluation that separates calibration from informativeness.

Confidence sets for ranks must accommodate population ties (Hall and Miller 2009; Mogstad et al. 2024). Combinatorial inference for graphical models addresses other population graph properties under different structural assumptions (Neykov et al. 2019); a correlation-optimal spanning tree is not a conditional-independence graph. In financial networks, Gałązka (2011) studied Polish stock correlations, while Tumminello et al. (2007); Musciotto et al. (2018) assessed bootstrap edge reliability. Our target is a simultaneous population degree or rank statement, rather than an edge selection frequency or a portfolio-performance claim.

The empirical distinction matters. In a separated factor model, direct comparisons can certify the hub even when the rectangular baseline is uninformative. Conversely, the Pearson-bootstrap calibration examined here still fails in some finite-sample designs. We retain those failures and interpret the Polish application as a diagnostic of resolution. Neither a valid optimization bound nor a high projected coverage rate repairs an invalid input calibration. A fixed annual sensitivity grid extends the

application to longer lookbacks and smaller nested liquid-stock groups, with forward diagnostics. These analyses improve degree resolution but still leave hub ranks unresolved; their distinction between persistent edge orders and changing estimated hubs is reported descriptively.

2 Population target and simultaneous orders

Let $G = (V, E)$ be a fixed connected graph with p vertices and m edges. For weights w, let

$$\mathcal{T}(w) = \arg \max_{T \text{ spanning tree}} \sum_{e \in T} w_e.$$

In the correlation application G is complete. A maximum-correlation tree has the same optimal-tree family as the minimum-distance tree based on $\sqrt{2(1 - w_e)}$: both are determined by the same edge ordering, including ties. Strictly increasing edge transformations preserve this family.

Write $S_i = \{e : i \in e\}$ and $d_i(T) = |T \cap S_i|$. Descending ranks of vertex i, including every resolution of equal degrees, range from

$$r_i^-(T) = 1 + \sum_{j \neq i} \mathbf{1}\{d_j(T) > d_i(T)\}, \qquad r_i^+(T) = p - \sum_{j \neq i} \mathbf{1}\{d_j(T) < d_i(T)\}. \tag{1}$$

The inferential target contains these quantities for every $T \in \mathcal{T}(w_0)$. Under complete independence all edges have weight zero, every labelled tree is optimal, and the appropriate degree and rank targets are $[1, p-1]$ and $[1, p]$. Selecting one arbitrary tied optimum would artificially narrow this target.

A random strict relation $\widehat{\succ}$ is simultaneously valid if

$$\mathbb{P}(\mathcal{E}) \geq 1 - \alpha, \qquad \mathcal{E} = \{e \mathrel{\widehat{\succ}} f \implies w_{0e} > w_{0f} \text{ for all } e, f\}. \tag{2}$$

We may take its transitive closure without an additional error budget. Every computational restriction below is conditional on this same event. The relation need not specify all pairwise edge comparisons, and its incompleteness is a source of conservative bounds.

3 An exact Kendall comparison construction

Suppose $X_1, \ldots, X_n$ are IID p-vectors. Coordinates may be dependent or discrete, and no moment assumption is imposed. For independent copies X, X', define

$$\tau_{ij} = \mathbb{E}[\operatorname{sgn}(X_i - X_i') \operatorname{sgn}(X_j - X_j')].$$

This is Kendall tau-a, not the tie-normalized tau-b. Attach independent uniform random variables U_{ti} to each observation and coordinate, and order $\widetilde{X}_{ti} = (X_{ti}, U_{ti})$ lexicographically. Randomization is independent of the observed sample.

For a predetermined pairing $(1,2),(3,4),\ldots$, let $N = \lfloor n/2 \rfloor$ and

$$S_{ki} = \operatorname{sgn}(\widetilde{X}_{2k-1,i} - \widetilde{X}_{2k,i}), \qquad H_{k,e} = S_{ki}S_{kj}, \quad e = \{i,j\}.$$

Here $H_{k,e} \in \{-1,1\}$ and the vectors H_k are IID. For an ordered comparison (e,f) set

$$A_{ef} = \sum_k \mathbf{1}\{H_{k,e} = 1, H_{k,f} = -1\},$$
$$B_{ef} = \sum_k \mathbf{1}\{H_{k,e} = -1, H_{k,f} = 1\}, \qquad D_{ef} = A_{ef} + B_{ef}.$$

The one-sided exact p-value is

$$p_{ef} = \mathbb{P}\{\mathrm{Bin}(D_{ef}, 1/2) \geq A_{ef}\}, \tag{3}$$

with value one when $D_{ef} = 0$. This is the conditional McNemar test (McNemar 1947), applied to paired edge kernels.

We test both directions of every pair of distinct edges sharing a vertex. On the complete graph this fixed family has $K = p(p-1)(p-2)$ hypotheses. Apply Holm's step-down rule (Holm 1979) at level α: ordered p-values $p_{(k)}$ are rejected until the first failure of $p_{(k)} \leq \alpha/(K-k+1)$. Rejections define $e \widehat{\succ} f$; subsequently take the transitive closure.

Theorem 1 (Finite-sample order and hub coverage) *For IID rows and the predetermined pairing above, the resulting relation satisfies (2) with $w_{0e} = \tau_e$, for every finite $n \geq 2$. The probability includes auxiliary tie randomization. Consequently, every degree, degree-difference and rank bound obtained by the valid outer projections in Sections 4–5 covers its target simultaneously with probability at least $1-\alpha$, for all population-optimal trees and every resolution of degree ties.*

Proof Conditional on X, X', if neither coordinate is tied, the augmented sign product equals the original sign product. If either is tied, the independent auxiliary sign for that coordinate has mean zero. Thus $\mathbb{E}H_{k,e} = \tau_e$, including when both coordinates tie. Put $\pi_+ = \mathbb{P}(H_e = 1, H_f = -1)$ and $\pi_- = \mathbb{P}(H_e = -1, H_f = 1)$. Then $\tau_e - \tau_f = 2(\pi_+ - \pi_-)$. Conditional on D_{ef}, A_{ef} is binomial with success probability $q = \pi_+/(\pi_+ + \pi_-)$ whenever the denominator is positive. Under $\tau_e \leq \tau_f$, $q \leq 1/2$, so (3) is superuniform by binomial stochastic monotonicity. The zero-denominator case gives $D_{ef} = 0$ almost surely. Holm's argument therefore controls the probability of any false strict order by α, regardless of dependence among comparisons. Transitive consequences remain true on the same event. Each subsequent outer projection retains all population-optimal trees on that event, proving the final assertion. □

The pairing cannot be selected after inspecting which arrangement gives favorable tests. A fixed pairing, or independent random pairing, is valid; trying several and retaining the best requires additional multiplicity control. This construction deliberately uses N independent pairs. The full-sample Kendall estimator uses all $n(n-1)/2$

pairs, whose kernels are dependent. Treating those kernels as independent would invalidate the binomial calculation. Applying the IID result to serially dependent financial returns or fitted residuals is also unjustified.

Proposition 2 (A sufficient discordance-dependent power bound) *Consider $s \geq 1$ selected true comparisons from the tested family. Suppose each has gap $\Delta = \tau_e - \tau_f \geq \delta > 0$ and discordance probability $d = \mathbb{P}(H_e \neq H_f) \leq \bar{d}$. If*

$$N \geq \frac{64\bar{d}}{\delta^2} \log \frac{2Ks}{\alpha\eta}, \qquad 0 < \eta < 1, \tag{4}$$

then all these comparisons are rejected with probability at least $1 - \eta$.

Proof Let $Z_k = (H_{k,e} - H_{k,f})/2$. Its mean is $\Delta/2$, variance is at most $\bar{d}$, and its centered absolute value is at most two. Bernstein's inequality gives

$$\mathbb{P}\{A - B < N\delta/4\} \leq \exp\{-N\delta^2/(32\bar{d} + 16\delta/3)\} \leq \exp\{-N\delta^2/(64\bar{d})\},$$

where $\delta \leq 2\bar{d}$. A second Bernstein bound gives $\mathbb{P}(D > 2N\bar{d}) \leq \exp(-3N\bar{d}/8)$. Outside these events, the binomial Hoeffding bound implies

$$p_{ef} \leq \exp\{-(A - B)^2/(2D)\} \leq \exp\{-N\delta^2/(64\bar{d})\}.$$

Condition (4) makes this p-value at most α/K. It also bounds each bad-event probability by $\alpha\eta/(2Ks)$, since $\delta^2 \leq 4\bar{d}^2$. A union bound over the s comparisons yields failure probability at most $\alpha\eta/K \leq \eta$. Every p-value at most α/K is rejected by Holm. □

The proposition is sufficient, deliberately conservative, and not a necessary sample-size threshold. Its role is to exhibit the discordance probability in the power mechanism. It does not establish uniform superiority over full-sample U-statistic procedures. In particular, the experiments below remain uninformative in many short-sample designs.

4 Projecting orders into joint degree information

4.1 Classical screening and exact extrema on an outer minor

For an edge $e = \{u, v\}$, exclude e if a u–v path consists entirely of edges certified above e. The cycle property rules it out of every population-optimal tree. Force e if u, v become disconnected after deleting e and all edges certified below e. The resulting cut has e strictly heavier than every alternative, so every optimum contains it. These are sufficient screening rules for our partial relation; we do not assert that they characterize all possible and necessary edges under general dependent uncertainty sets.

Let A be the allowed edges and F the forced edges. On $\mathcal{E}$, F is a forest. Contract F, delete excluded edges, and retain the resulting graphic matroid M with rank $k = p-1-|F|$ and rank function r. Parallel edges and loops are permitted after contraction. Every population-optimal tree is $F \cup B$ for some base B of M, but many bases can fail to be optimal for any compatible weights. All exactness claims below refer to this outer base family.

Write $a_i = |F \cap S_i|$ and $I_i = S_i \setminus F$ restricted to the residual ground set. The ordinary greedy rank identities give

$$\underline{d}_i = a_i + k - r(E_M \setminus I_i), \qquad \overline{d}_i = a_i + r(I_i). \tag{5}$$

For vertices i, j, assign residual coefficients $c_e = \mathbf{1}\{e \in S_i\} - \mathbf{1}\{e \in S_j\}$, and let P_+, P_0, P_- denote their level sets.

Proposition 3 (Degree-difference extrema on the minor) *The exact maximum and minimum of $d_i - d_j$ over the outer base family are*

$$U_{ij} = a_i - a_j + r(P_+) + r(P_+ \cup P_0) - k, \tag{6}$$
$$L_{ij} = a_i - a_j + k - r(P_-) - r(P_- \cup P_0). \tag{7}$$

Proof For a base B of size k, $\sum_{e \in B} c_e = |B \cap P_+| + |B \cap (P_+ \cup P_0)| - k$. Each intersection is bounded by the corresponding rank. A greedy base processing P_+ first, then P_0, then P_- attains both bounds simultaneously, by the matroid augmentation property. This proves (6). Reverse signs for (7). The same argument with coefficients zero and one gives (5). □

These are consequences of classical matroid optimization, not new exchange principles. Their inferential use is that $L_{ij} > 0$ certifies $d_i > d_j$ for every retained tree. Directly optimizing a difference retains a common feasible tree, whereas subtracting a marginal maximum from a marginal minimum can use incompatible trees.

4.2 Cycle-path incompatibilities

Individual screening can leave many jointly impossible edge configurations. Suppose P is a simple path between the endpoints of e, and $e \widehat{\succ} f$ for some $f \in P$. Every population-optimal tree satisfies

$$\sum_{g \in P} x_g \leq |P| - 1, \tag{8}$$

where x_g is its edge indicator. Indeed, a tree containing all of P cannot also contain e; replacing f by e would then strictly increase its weight. Notice that the condition needs one lighter edge on the path. It is stronger than excluding only an edge that is lighter than an entire alternative path.

We use every such path of length two and additional paths from eight candidate spanning trees. Candidate priorities are generated with a fixed seed, with forced edges processed first and excluded edges omitted. For each candidate and each non-tree edge, its unique path is checked against the same certified relation. Path selection may depend on the relation without a new statistical correction: every inserted inequality is valid on $\mathcal{E}$. Eight is a computational design choice, not a claim that these paths exhaust the useful constraints.

The linear relaxation has $0 \leq x_e \leq 1$, $\sum_e x_e = p - 1$, $\sum_{e \in S_i} x_e \geq 1$, fixed forced/excluded coordinates, and the selected inequalities (8). To impose connectivity,

add directed nonnegative flows, root outflow minus inflow $p-1$, and outflow minus inflow -1 at every other vertex. For an undirected edge $e=\{u,v\}$ impose

$$f_{uv}+f_{vu}\le (p-1)x_e.$$

Every tree admits these flows by routing one unit from the root to each other vertex. Fractional solutions may remain, so this is an outer relaxation, not the convex hull of compatible optimal trees.

We minimize and maximize every degree, and optimize both directed degree differences between the observed highest-degree vertex and every other vertex. This anchor can be selected from the data because all projected statements use the common event $\mathcal{E}$. Ties in anchor selection are resolved by the frozen vertex order. The minor bounds remain available for every other pair.

4.3 Rational certificates for numerical optimization

A floating-point optimum should not be rounded into an inferential statement without a valid bound. Write the relaxation as $Ax\le b$, $Ex=h$, $\ell\le x\le u$, incorporating both edge and flow variables. Its coefficients and bounds are integers. For an integer objective c, take any rational multipliers $y\le 0$ and unrestricted z, and put $v=c-A^\top y-E^\top z$.

Proposition 4 (Verifiable objective bound) *For every feasible x,*

$$c^\top x\ge b^\top y+h^\top z+\sum_j \min\{v_j\ell_j, v_j u_j\} =: q. \tag{9}$$

If the target objective is integer, $\lceil q\rceil$ is a valid lower bound. Applying the formula to $-c$ gives upper bound $\lfloor -q\rfloor$.

Proof Expand $c^\top x=y^\top Ax+z^\top Ex+v^\top x$. The first term is at least $y^\top b$ because $y\le 0$; the second equals $z^\top h$; coordinatewise minimization over the box bounds the last term. Integer rounding preserves the assertion. □

The implementation takes inequality multipliers from HiGHS, clips them to be nonpositive and rounds downward to multiples of 2^{-20}. Equality multipliers are rounded to that denominator. The residual and bound are then computed with checked integer arithmetic. Saved sparse certificates contain the numerator, denominator and multipliers for every improvement. An independent verifier reconstructs the constraints and recomputes the bound with Python arbitrary-precision integers. Validity does not require that the numerical solution satisfy KKT equations exactly, or even that its proposed multipliers be optimal.

5 Rank projection with degree conservation

Let integer degree bounds be $L_i \le d_i \le U_i$, and impose

$$\sum_i d_i = 2p - 2 =: D_0, \qquad d_i \ge 1. \tag{10}$$

We compute sharp rank endpoints over this integer box intersected with (10). For vertex i, enumerate feasible $s = d_i \in [L_i, U_i]$, so that $\sum_{j\neq i} L_j \le D_0 - s \le \sum_{j\neq i} U_j$.

For the worst rank, all vertices with $L_j \ge s$ are already tied with or above i. Each optional vertex with $L_j < s \le U_j$ costs $s - L_j$ additional degree units to reach that threshold. Starting from all lower bounds, select the cheapest such costs within budget $D_0 - s - \sum_{j\neq i} L_j$. One plus the mandatory and selected counts is the largest possible rank for that s.

For the best rank, vertices with $L_j > s$ necessarily outrank i. Without raising any additional vertex above s, the largest possible total for the other degrees is

$$C(s) = \sum_{j:L_j>s} U_j + \sum_{j:L_j\le s} \min(U_j, s).$$

If $D_0 - s > C(s)$, select the fewest additional vertices whose capacities $U_j - s > 0$ cover the shortfall, taking largest capacities first. Add their count and the mandatory count to one. Minimize this best rank and maximize the worst rank over feasible s.

Proposition 5 (Sharpness within the degree relaxation) *The threshold and capacity calculations above attain the best and worst ranks over all integer vectors satisfying the degree bounds and (10). On the complete graph every positive integer vector with this total is the degree sequence of a labelled tree. Sharpness need not hold for the smaller family of order-compatible or minor-compatible trees.*

Proof To make k optional competitors reach s, their required increases have minimum total equal to the sum of the k smallest costs. The budget criterion is therefore necessary and sufficient; remaining degree units can be allocated within the original upper bounds. If this forced another optional competitor to reach s, that competitor would already have been affordable, contradicting maximality. For the best rank, activating a vertex above s supplies at most $U_j - s$ additional units. The largest-capacity rule minimizes the number required. Feasibility of s guarantees enough total capacity, and integer intervals allow filling the remaining total without gaps. Finally, a Prüfer sequence containing vertex i exactly $d_i - 1$ times realizes any positive vector summing to $2p - 2$. □

Intersect these rank intervals with direct comparison bounds

$$1 + \#\{j : L_{ji} > 0\} \ \le r_i \le \ p - \#\{j : L_{ij} > 0\}. \tag{11}$$

A certified unique hub has upper rank one. A vertex remains a possible hub whenever its lower rank is one. Broad intervals are a legitimate outcome; we do not resolve

their ties using estimated degrees. The proof of Theorem 1 now follows through every computational stage.

6 A positive-definite example of lost joint information

Consider eight vertices indexed $0, \dots, 7$. In the block $\{0, \dots, 5\}$ let hub correlations be $\rho_{0j} = 0.06 + t$ for $1 \le j \le 5$, and leaf correlations $\rho_{ij} = 0.04 + t$ for $1 \le i < j \le 5$. Set $\rho_{06} = 0.10$, $\rho_{67} = 0.15$, all other cross-block correlations to zero, and diagonal entries to one, with $t \in [-0.025, 0.025]$.

Every matrix is positive definite: it is symmetric, has positive diagonal and is strictly diagonally dominant; the largest absolute off-diagonal row sum is 0.525. Every within-block spoke exceeds every within-block leaf edge by 0.02. The unique optimal tree throughout the curve consists of five spokes, the bridge 06 and edge 67. Its degree vector is $(6, 1, 1, 1, 1, 1, 2, 1)$, so vertex zero is uniquely first.

Now transform correlations by Fisher's atanh and subtract the mean across all 28 edges at each t. Form the smallest coordinatewise rectangular hull of this centered curve. Global centering removes a common shift across all edges but not a shift restricted to one block. Exact rectangular projection yields degree ranges

$$[2, 6],\ [1, 5] \text{ (each of five leaves)},\ [2, 2],\ [1, 1].$$

The marginal rank construction gives vertex zero $[1, 7]$. In contrast, the true local orders along the coupled curve force all seven tree edges and rank $[1, 1]$. For a changing edge, the derivative of its centered Fisher weight is bounded below by $1 - 15/[28(1 - 0.085^2)] > 0$; for a constant edge it is negative. Thus its coordinate extrema occur at the two endpoints. An executable verification is supplied with the code. This is a deterministic comparison of uncertainty-set shapes, not a claim that this curve is an estimated confidence region. It shows why rectangularization can discard relevant information even when the candidate matrices are positive definite and global centering has already been used.

7 Monte Carlo evaluation

7.1 Design and estimands

All experiments use nominal $\alpha = 0.05$, fixed seeds recorded before the production runs, and development-pilot seeds excluded from those runs. This is retrospective method development, not a preregistered study. We retain all calibration failures. Finite-sample experiments use 200 replications per cell; bootstrap experiments use 100. The supplement supplies every replication and full cell summaries. Monte Carlo uncertainty is reported using Wilson intervals for selected proportions.

In the separated Gaussian model, $X_i = a_i Z + \sqrt{1 - a_i^2}\epsilon_i$, with independent standard normal factors, $a_0 = 0.99$ and remaining loadings equally spaced from 0.45 to 0.55. In the near-tie model $a_0 = 0.8 + 0.5/\sqrt{n}$, $a_1 = 0.8$ and the others span $[0.45, 0.55]$. The t_5 model divides each Gaussian row by an independent $\sqrt{\chi_5^2/5}$. These non-null models have a unique population star centred at vertex zero. For elliptical

Gaussian and t distributions, population Kendall weights are $2\arcsin(\rho)/\pi$. Gaussian independence and a discrete IID uniform distribution on $\{-1, 0, 1\}$ assess exact ties.

The finite-sample experiment has $p = 12$, $n \in \{252, 1008, 4096\}$ for four continuous models, and $p = 52$, $n \in \{252, 4096\}$ for the same models. Two additional discrete cells use $p = 12$, $n \in \{252, 1008\}$. The resulting 22 cells contain 4,400 data sets. We compare the exact order construction followed by minor and conservation projection with a full-sample Kendall tau-a rectangular Hoeffding band. The latter uses

$$\varepsilon = \sqrt{\frac{2\log(2m/\alpha)}{\lfloor n/2 \rfloor}}, \qquad [L_e, U_e] = [\widehat{\tau}_e - \varepsilon, \widehat{\tau}_e + \varepsilon] \cap [-1, 1].$$

Its projection uses exact per-vertex rectangular degree endpoints and marginal rank envelopes. This deliberately compares the earlier full-sample baseline with the disjoint-pair comparison procedure; the computational projection also differs. We do not attribute all gains to one component. Pairwise and conservation bounds are separately implemented and tested. The expensive path LP is evaluated on the complete empirical panel and exhaustive small examples, rather than being included in every Monte Carlo replication.

Coverage requires simultaneous inclusion of all true degree and rank targets. For null models we check the entire tied-tree target, not a selected tree. We also measure any false certified edge order, mean interval width, the number of possible hubs, the probability of certifying the true hub, and the number of forced edges. Hub certification is distinct from complete-tree recovery. Tied leaves of a known star still have rank range $[2, p]$.

Across the finite-sample cells, the smallest observed simultaneous degree coverage was 0.995, and the smallest rank coverage was 1.000. The largest observed false-order frequency was 0.070 (14/200), in the $p = 12$, $n = 4096$ Independence cell; its Wilson 95% interval is [0.042, 0.114]. A Monte Carlo proportion above 0.05 is not itself a contradiction of the finite-sample theorem. We report it rather than truncating observed errors at the nominal level.

Figure 1 distinguishes hub certification from complete-tree recovery in the separated models. The full-sample rectangular baseline certified no hub at the displayed sample size. Successful hub inference need not imply that every edge is forced. Conversely, the short-sample and near-tie cells largely remain unresolved. A condition based on four uniform-band half-widths exceeding a tree separation margin is a sufficient recovery condition; failure of that condition is not an impossibility result for all Kendall methods.

7.2 Pearson-bootstrap calibration as a separate diagnostic

The bootstrap branch uses all rows and local differences of Fisher-transformed Pearson correlations. For each local pair (e, f), let $\widehat{\theta}_{ef} = \operatorname{atanh}(\widehat{\rho}_e) - \operatorname{atanh}(\widehat{\rho}_f)$. With bootstrap

Table 1 Finite-sample experiment: 200 replications per cell. FWER is the frequency of any false strict order. Hub columns report certification of vertex zero as uniquely first; null cells have no unique population hub.

p	Model	n	Order FWER	Hub: order	Hub: box
12	Discrete null	252	0.025	—	—
12	Discrete null	1008	0.030	—	—
12	Gaussian star	252	0.020	0.000	0.000
12	Gaussian star	1008	0.010	0.000	0.000
12	Gaussian star	4096	0.000	1.000	0.000
12	Independence	252	0.000	—	—
12	Independence	1008	0.030	—	—
12	Independence	4096	0.070	—	—
12	Near tie	252	0.005	0.000	0.000
12	Near tie	1008	0.000	0.000	0.000
12	Near tie	4096	0.000	0.000	0.000
12	t_5 star	252	0.000	0.000	0.000
12	t_5 star	1008	0.010	0.000	0.000
12	t_5 star	4096	0.000	1.000	0.000
52	Gaussian star	252	0.000	0.000	0.000
52	Gaussian star	4096	0.000	0.790	0.000
52	Independence	252	0.035	—	—
52	Independence	4096	0.035	—	—
52	Near tie	252	0.000	0.000	0.000
52	Near tie	4096	0.005	0.000	0.000
52	t_5 star	252	0.015	0.000	0.000
52	t_5 star	4096	0.010	0.780	0.000

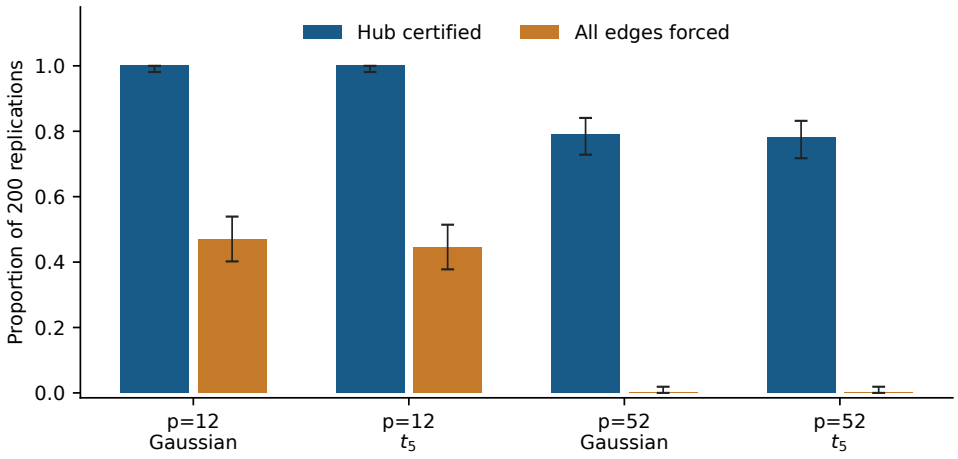


Fig. 1 Certification at $n = 4096$, from 200 replications per cell. Hub certification uses the upper rank endpoint; full recovery requires all $p - 1$ edges to be forced by screening. Whiskers are Wilson 95% Monte Carlo intervals. The Hoeffding-box hub rate is zero in all four cells.

standard deviation s_{ef} held fixed across resamples, define

$$Q_b = \max_{(e,f)} \left| \frac{\widehat{\theta}_{ef}^{*(b)} - \widehat{\theta}_{ef}}{s_{ef}} \right|.$$

The empirical upper 0.95 quantile gives simultaneous nominal intervals $\widehat{\theta}_{ef} \pm q s_{ef}$; only intervals excluding zero create strict orders. We reject degenerate studentizers rather than replacing them silently. Contrast calibration cancels shared shifts, but that algebraic property does not guarantee finite-sample coverage.

There are 24 cells: $p \in \{12, 52\}$, $n \in \{252, 1008\}$, and Gaussian independence, separated star, near tie, t_5 star, and two Gaussian AR(1) versions (independence and star) with coefficient 0.4. Stationary Gaussian innovations preserve the contemporaneous correlation target; burn-in is discarded. Each data set uses 499 bootstrap resamples. IID cells resample rows; AR cells use circular blocks of length $\max\{2, \text{round}(n^{1/3})\}$. The same resamples calibrate the globally centered Fisher-z rectangular comparator. We separately report simultaneous input coverage, false-order familywise error and projected coverage.

Table 2 Pearson-bootstrap calibration at $p = 52$: 100 data sets and 499 resamples per cell. Coverage concerns the whole input region, not a selected edge. Full results for $p = 12$ are supplied with the replication files.

Model	n	Box cover	Contrast cover	Order FWER
AR null	252	0.68	0.22	0.78
AR null	1008	0.83	0.78	0.22
AR star	252	0.67	0.26	0.36
AR star	1008	0.86	0.76	0.03
Gaussian star	252	0.90	0.78	0.06
Gaussian star	1008	0.93	0.93	0.01
Independence	252	0.86	0.82	0.18
Independence	1008	0.96	0.90	0.10
Near tie	252	0.94	0.85	0.03
Near tie	1008	0.88	0.88	0.01
t_5 star	252	0.74	0.65	0.18
t_5 star	1008	0.83	0.79	0.03

The smallest contrast-region coverage is 0.22, in the $p = 52$, $n = 252$ AR null cell (Wilson interval [0.150, 0.311]). For Gaussian independence at $p = 52, n = 252$, contrast coverage is 0.82 and false-order frequency is 0.18. Under the exact null, every false strict order is a calibration error even if the downstream tree family remains broad enough to cover every degree. Projected degree coverage therefore cannot be used as a substitute for input calibration.

The block bootstrap is motivated by the dependent-resampling literature (Künsch 1989), but no theorem in this paper validates this particular growing collection of

studentized contrasts for financial returns. The observed failures preclude labelling the empirical bootstrap intervals as established 95% confidence intervals. Increasing the number of bootstrap draws would reduce simulation noise in its quantile estimate; it would not by itself establish the missing calibration guarantee.

8 Polish equity reanalysis

We reuse a frozen Polish-market data assembly underlying the preceding version of this study (Gałązka and Wdowicka 2026). The historical motivation is Gałązka (2011); the current data do not reproduce the 2011 sample. The assembly combines publicly available Bankier price histories, the WIG series and archived GPW index-portfolio information. It retains the previously recorded eligibility lists, source checks and data hashes. Thus changes in inference are not confounded with reselecting a more favorable stock universe.

There are 231 historical origins from 2007–2025. At each origin we use 252 complete daily close-to-close log returns for the recorded eligible stocks, with 41–60 vertices across windows. Of 171 recorded security identities, histories were available for 164; 155 appear in a retained window. Eligibility requires complete past returns and volumes, positive volume on at least 90% of sessions, at most 15% zero close-to-close returns and positive return variance. No unavailable quotation is imputed. Membership follows archived quarterly portfolios rather than a fully reconciled daily record of extraordinary replacements. Prices are used as supplied and are not certified cash-dividend-inclusive total returns. Missing-history exclusions and corporate-action limitations remain those documented in the frozen source audit. These are available historical series, not a newly acquired complete licensed GPW database; survivorship and vendor adjustment limitations cannot be ruled out merely by the eligibility check.

We analyse raw returns, daily cross-sectional demeaning, and residualization on WIG. All transformations are recomputed from the original and bootstrap covariance matrices, including factor fitting within resamples. Each origin uses 999 circular block-bootstrap replicates with block length ten; all three representations share the resampled covariance draws. This is conditional analysis for each frozen window, not simultaneous coverage across 231 overlapping origins or across representations. The exact IID theorem is not invoked for these returns.

For each of the 693 origin–representation combinations, we perform local contrast calibration, cut/cycle screening, minor degree-difference projection, all selected path constraints, rationally certified LP degree bounds, and anchor-to-all degree comparisons. The anchor is the largest observed tree degree, with frozen-list tie handling. We then intersect conservation and direct-comparison rank bounds. The empirical evaluation therefore applies the stronger projection to the whole frozen panel rather than reporting a selected favorable window.

The refined projection reduces the raw-network mean degree width from 49.89 to 45.99, a 7.8% reduction relative to the minor. It does not resolve hub ranks: all 35,979 reported vertex–window–representation rank intervals remain $[1, p]$. No unique hub and no top-decile membership is certified. There are 85,129 saved rational certificates

Table 3 All 231 windows per representation. Widths are averages of the within-window mean degree width. Full ranks is the mean fraction of vertices with interval $[1, p]$. Hubs counts certified unique hubs over all windows. These are nominal bootstrap diagnostics.

Representation	Minor width	Path-LP width	Full ranks	Hubs
Raw	49.89	45.99	1.00	0
Demeaned	49.90	46.51	1.00	0
WIG residual	49.91	47.63	1.00	0

for improved degree or anchor-contrast endpoints. These improvements demonstrate a stricter feasible-set projection, but not an informative financial hub ranking.

The all-ranks result is specific to these windows, this calibration and these outer projections. It does not prove that no population hub exists or that no other statistically justified procedure could identify one. It also does not validate the nominal error rate: the calibration failures in Table 2 remain relevant.

8.1 Longer windows, nested liquid universes and forward diagnostics

The full-panel result motivates a sensitivity analysis that separates sample length from changing stock composition. Before computing the extended outcomes, we fixed the last recorded origin in every represented calendar year, giving 19 annual origins. This is a retrospective sensitivity grid, not a preregistration or an independent replication of method development. At each origin, we use the longest calendar-available lookback from 504 and 1008 sessions to screen the original eligible list. Eligibility requires complete past returns, supplied prices and volumes, positive volume on at least 90% of sessions, at most 15% zero returns, and positive variance. Within this eligible set we rank stocks by median supplied closing price times volume, breaking ties by ISIN. This is an available-data liquidity proxy, not a certified corporate-action-adjusted traded-value measure.

We take nested groups of 10, 20 and 40 stocks. Their membership is kept fixed when comparing 252, 504 and 1008 sessions at the same origin. Changing group size changes the population tree target; a hub of a small liquid subnetwork need not be a hub of the full market. Demeaning is recomputed within each selected group, while factor residualization retains the same WIG series. All dated observations used for selection precede or coincide with the origin; future observations never enter selection. Vendor histories can incorporate later revisions, so this rule does not establish a fully point-in-time data source.

Of 513 planned origin–lookback–group–representation combinations, 81 cannot be evaluated because of insufficient calendar history or too few eligible stocks. We retain the full exclusion list. The remaining 432 primary analyses use block length ten and 999 common-row resamples per origin and lookback, across all three representations. A further 34 raw-network analyses examine block lengths five and twenty at $p = 20, n = 1008$, for all 17 available origins. These are nominal, per-configuration

bootstrap projections. We do not claim simultaneous coverage over the grid or valid 95% coverage for dependent returns.

Table 4 Raw-network comparison on identical stocks and matched origins. The 1008-session availability criterion leaves 17 origins for groups of 10 and 20, and ten for groups of 40. Widths and forced-edge counts are within-group origin averages. Full ranks is the fraction of vertex ranges remaining $[1, p]$.

p	Origins	n	Degree width	Forced edges	Full ranks
10	17	252	6.71	0.47	1.00
10	17	504	6.18	0.59	1.00
10	17	1008	5.78	1.00	1.00
20	17	252	15.04	0.47	1.00
20	17	504	14.14	0.65	1.00
20	17	1008	12.92	1.35	1.00
40	10	252	32.82	0.00	1.00
40	10	504	31.03	0.20	1.00
40	10	1008	29.45	0.80	1.00

The matched-origin mean degree width decreases by 13.9%, 14.1% and 10.3% when moving from 252 to 1008 observations for groups of 10, 20 and 40 respectively. More edges are forced under the nominal order region, but none of the 432 primary or 34 block-sensitivity configurations yields a non-full vertex rank interval or a certified unique hub. The negative ranking result therefore survives a fourfold lookback extension, liquidity-based dimension reduction, and the stated block checks. It remains a result about this procedure and these targets, not an impossibility theorem.

Table 5 Block-length sensitivity for the same 17 raw-network origins, $p = 20, n = 1008$. Resampling seeds differ between block lengths.

Block length	Origins	Degree width	Forced edges	Full ranks
5	17	12.94	1.24	1.00
10	17	12.92	1.35	1.00
20	17	12.93	1.29	1.00

For a forward diagnostic, we hold the training stock group fixed and estimate a new tree using the next 252 sessions. The diagnostic is available only when every selected stock and the factor have complete future returns. We neither impute unavailable histories nor drop a future-missing stock from the selected group. There are 93 complete primary configurations per representation, out of 144; the remaining 51 have incomplete future histories or insufficient follow-up. Overlapping lookbacks and nested

groups mean these are not 93 independent samples. Forward summaries are conditional on future-history availability and may be affected by attrition; no reweighting corrects that selection. We compute the fraction of transitive training order statements whose direction agrees with the future correlation estimates, and whether the training anchor remains tied for largest future degree. These are descriptive temporal-stability measures, not population coverage tests or portfolio-performance results.

Table 6 Forward diagnostics over the next 252 sessions. Order agreement averages the within-configuration fractions, giving each configuration equal weight. Anchor still top includes ties and does not establish a unique hub.

Representation	Configurations	Order agreement	Anchor still top
Raw	93	95.7%	35/93
Demeaned	93	95.2%	19/93
WIG residual	93	93.9%	19/93

Estimated edge-order directions persist much more frequently than the identity of the estimated highest-degree vertex. This comparison involves different functionals: preservation of many inequalities is not equivalent to preservation of a hub, and a high fraction of correct directions need not mean that every direction is preserved jointly. Because the population can change over time, the forward sample is not a known-truth coverage benchmark. The calibration failures in Section 7 remain unresolved.

As an illustration, the earliest complete raw-network configuration with anchor bootstrap frequency at least 0.9 and loss of top position in the future sample is the 2008 annual origin, $p = 10, n = 504$. The anchor is PKOBP; it is tied for highest degree in 91.2% of bootstrap trees, while its nominal degree range is [1, 9] and rank range is [1, 10]. Its future estimated rank range is [3, 10], despite 100.0% directional agreement for the certified training orders. The illustration is selected retrospectively by the stated rule; all configurations are reported, and it is not additional confirmatory evidence.

Independent verification reconstructs the selection rule for all 50 origin–group universes, checks five source-input hashes and all 9,710 additional vertex records, recomputes the 303 available forward analyses including block sensitivities, and validates all 24,419 improving rational LP certificates. Every check passes. Longer financial windows increase sample size while raising the possibility of regime changes, so their improved degree resolution must be interpreted alongside that stationarity limitation.

9 Computational verification and limitations

The implementation checks 300 random graphic minors against exhaustive enumeration of all labelled trees for $p = 3, 4, 5, 6$. All individual degree and pairwise degree extrema agree. The same 300 instances check exact rank projection over every feasible

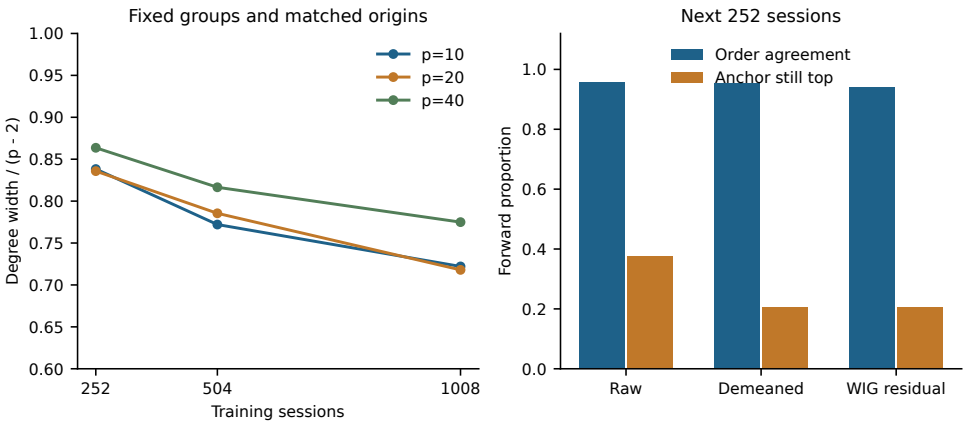


Fig. 2 Extended empirical diagnostics. Left: mean raw-network degree width divided by $p - 2$, on the matched origins and fixed stock groups in Table 4. Right: forward edge-order agreement and retention of the training anchor among the highest-degree vertices, from Table 6. Both panels are descriptive; neither supplies a validated confidence level for GPW returns.

degree vector. In 32 additional examples with tied integer edge weights, exhaustive optimal trees satisfy every inserted path constraint and every reported LP bound. Independent rational arithmetic verifies 242 improved small-instance certificates. Thirty tied and continuous samples check the accelerated full-sample Kendall tau-a against the SciPy-based reference, and 30 samples check the optimized discordance counts against direct indicator counts. Exact-binomial superuniformity is also checked numerically over 196 combinations of sample size, null probability and significance threshold. These checks support implementation correctness; the coverage claim rests on the proofs, not on the checks.

For the empirical run, independent Python-integer verification passed for all 85,129 saved improving certificates in all 693 origin–representation combinations. Source-input hashes, fixed seeds, full replication records, derived order relations and the verifier are retained. No raw third-party price series is required to check the saved optimization certificates.

The procedure still has limitations. Disjoint pairing sacrifices information, the local comparison family has $O(p^3)$ tests, and a weak or nearly tied hub may remain unidentified. The power bound is sufficient rather than sharp. A different valid joint test can be substituted, but its coverage must be established separately. The partial-order minor and the path LP are outer approximations; their integer degree bounds need not be sharp for the original statistical uncertainty set. The conservation rank endpoints are sharp only for their stated relaxation.

The implementation forms local kernel Gram matrices, computes transitive closure, and uses union–find rank evaluations for all vertex pairs. A straightforward minor implementation needs $O(p^2 m)$ edge visits up to union–find factors. The LP has $3m$ variables plus the selected path inequalities; the empirical specification solves

$2p + 2(p - 1)$ objectives per window. More path inequalities can improve the projection but increase computation. Rational certificates make numerical verification reproducible and do not change the uncertainty model.

10 Discussion

The inferential question is whether the data establish a degree comparison or hub rank across all population-optimal trees. Direct Kendall kernel comparisons provide one finite-sample answer under IID sampling. Joint tree restrictions then translate valid orders into stronger degree information, while the sum of tree degrees constrains ranks. The construction uses established testing and optimization facts, with explicit assumptions and a checked path from tests to the reported endpoints.

The favorable simulation cases show that a hub need not await recovery of every edge. The short-sample and Polish-market cases delimit this result: more detailed projection does not ensure a resolved rank, and bootstrap calibration remains a separate obstacle. In the additional empirical grid, retaining identical stocks across lookbacks allows a controlled comparison of resolution: 1008 sessions reduce mean degree width by about 10–14% relative to 252 sessions, without resolving ranks. Forward estimated edge-order agreement averages about 94–96%, whereas retention of the estimated hub among tied top vertices is much less frequent. These descriptive comparisons concern different functionals and changing populations; they provide temporal-stability evidence rather than a coverage guarantee. Further progress for dependent financial data requires a justified simultaneous comparison procedure or additional defensible structure, rather than interpreting uncalibrated bootstrap frequencies as population confidence.

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Data availability. Derived results, fixed simulation designs, data provenance and verification code accompany the article. The empirical analysis uses the frozen historical data assembly described above. Raw third-party price histories are not redistributed in the public code repository; original source records and hashes document the inputs and their limitations.

Code availability. The implementation and reproducibility materials are supplied in the accompanying revision package. The project repository is https://github.com/mgalazka84/gpw-hub-reliability, branch `empirical-revision-v0.4`, directory `contrast_inference`. Earlier releases are retained for comparison.

Use of generative AI. Generative AI assisted with coding, data handling, methodological development and manuscript drafting. The authors are responsible for checking the mathematical arguments, computations, references and final text.

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